

Data Science Practicum II, Proposal

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Predicting insurance claim costs
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Abstract

This project applies machine learning to predict insurance claim costs using the Allstate Claims Severity dataset from Kaggle. The dataset contains 188,318 insurance claims with categorical and numerical features, and the target variable, *loss*, represents the dollar amount associated with each claim. The business goal is to help insurance claims managers identify potentially high-cost claims earlier, improve reserve planning, prioritize claim reviews, and allocate resources more effectively.

Exploratory data analysis showed that claim costs are strongly right-skewed. Approximately 83.59% of claims are below \$5,000, while only 23 out of 188,318 claims exceed \$40,000. This indicates that extremely high-cost claims are rare but may create greater financial risk.

Three regression models were evaluated: Linear Regression, Random Forest, and Gradient Boosting. Before tuning, Random Forest achieved the lowest RMSE of \$1,903.31, compared with \$1,945.67 for Gradient Boosting and \$2,016.61 for Linear Regression. After hyperparameter tuning, Gradient Boosting achieved the best overall performance, with an RMSE of \$1,884.82, an MAE of \$1,218.61, and an R^2 of 0.5646. Based on these results, the tuned Gradient Boosting model was selected as the final model for predicting and ranking claims by expected cost.

The recommended approach is to use the model as a decision-support tool and conduct an initial pilot in which the top 5% of model-ranked claims are routed to experienced adjusters for early review before reserves are finalized. In the held-out test set, the top 5% captured all 6 claims with actual losses above \$40,000. Although this result is encouraging, only 6 such extreme claims were present in the test set. Therefore, the 5% threshold should be validated on additional or future data before full operational deployment.

I. INTRODUCTION/BACKGROUND

Insurance companies process many claims, and the final cost of each claim can vary widely. Identifying potentially expensive claims early is important because it can help claims managers prioritize reviews, allocate resources more effectively, and support better reserve planning. This project focuses on using machine learning to predict insurance claim costs and provide decision support to insurance claims professionals. The target variable, *loss*, represents the dollar amount associated with each insurance claim.

The data used in this project comes from the Allstate Claims Severity competition published on Kaggle in 2016. The dataset contains 188,318 claims and 132 columns, including 116 categorical features, 14 numerical features, an ID column, and the target variable *loss*. The data was imported into Python and analyzed using tools such as *pandas* and *scikit-learn*. Initial data inspection found no duplicate rows or missing values. Because the target variable is a continuous dollar amount, regression analysis was selected as the appropriate modeling approach.

Exploratory data analysis showed that claim costs are strongly right-skewed. Most claims are relatively inexpensive, while a small number have extremely high costs. Approximately 83.59% of claims are below \$5,000, while only 23 out of 188,318 claims exceed \$40,000. Although these high-cost claims are rare, they may create greater financial risk. Identifying potentially expensive claims earlier may allow claims managers to review higher-risk cases sooner and make better-informed decisions when planning reserves.

Three regression models were evaluated: Linear Regression, Random Forest, and Gradient Boosting. Before tuning, Random Forest achieved the lowest RMSE of \$1,903.31, compared with \$1,945.67 for Gradient Boosting and \$2,016.61 for Linear Regression. After hyperparameter tuning, Gradient Boosting

achieved the best overall performance, with an RMSE of \$1,884.82, an MAE of \$1,218.61, and an R^2 of 0.5646. Based on these results, the tuned Gradient Boosting model was selected as the final model.

The model is intended to support claims professionals rather than replace their professional judgment. The dataset contains anonymized features and does not include claim dates, detailed policy information, or external factors. These limitations reduce business interpretability and may limit the model's ability to generalize to new insurance environments.

Therefore, the recommended approach is to use the tuned Gradient Boosting model to rank claims by predicted cost and initially route the top 5% of model-ranked claims to experienced adjusters for early review before reserves are finalized. In the held-out test set, the top 5% captured all 6 claims with actual losses above \$40,000. This result is encouraging; however, because only 6 such extreme claims were present in the test set, the 5% threshold should be validated on additional or future data before full operational deployment.

II. PROBLEM STATEMENT

Insurance companies need to estimate claim costs accurately to establish appropriate reserves and manage financial resources effectively. However, claim costs can vary widely, and a small number of claims can become extremely expensive. In the Allstate Claims Severity dataset, approximately 83.59% of claims are below \$5,000, while only 23 out of 188,318 claims exceed \$40,000. This shows that very high-cost claims are rare but may create greater financial risk for an insurance company.

The main problem addressed in this project is how machine learning can be used to predict insurance claim costs and identify potentially expensive claims early enough to support claims-management decisions. The target variable, *loss*, represents the dollar amount associated with each insurance claim. Predicting claim costs can help claims managers prioritize claim reviews, allocate resources more effectively, and support better reserve planning.

Predicting claim costs is challenging because the distribution of *loss* is strongly right-skewed. Most claims have relatively low costs, while a small number have extremely high costs. In addition, the dataset contains anonymized predictors and does not include claim dates, detailed policy information, or external factors. These limitations reduce the interpretability of individual predictors and make rare, high-cost claims more difficult to predict accurately. Therefore, the model is intended to serve as a decision-support tool for claims professionals rather than replace their professional judgment.

To address this problem, three regression models were evaluated: Linear Regression, Random Forest, and Gradient Boosting. Their predictive performance was compared using Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and the coefficient of determination (R^2). After model comparison and tuning, the best-performing model was selected to predict claim costs and rank claims according to their expected cost.

The final business objective is to use the selected model as a decision-support tool and conduct an initial pilot in which the top 5% of model-ranked claims are routed to experienced adjusters for early review before reserves are finalized. In the held-out test set, the top 5% captured all 6 claims with actual losses above \$40,000. This result is encouraging; however, because only 6 such extreme claims were present in the test set, the 5% threshold should be validated on additional or future data before full operational deployment.

III. RELATED WORK

Machine learning provides several approaches for predicting continuous outcomes such as insurance claim costs. Regression and tree-based ensemble methods are commonly used for this type of prediction because they can model relationships between input features and a continuous target variable. In this project, Linear Regression, Random Forest, and Gradient Boosting were evaluated to determine which approach could best predict the *loss* variable in the Allstate Claims Severity dataset. The dataset, published through Kaggle, contains 188,318 claims with 116 categorical and 14 numerical features used for prediction.

Linear Regression was selected as the baseline model because the target variable is a continuous dollar amount. A baseline model provides a simple reference point for evaluating whether more advanced machine-learning methods improve predictive performance. The Linear Regression model achieved an RMSE of \$2,016.61 and an R^2 of 0.5016.

Random Forest and Gradient Boosting were also evaluated because tree-based ensemble methods can capture more complex and nonlinear relationships in the data. Before hyperparameter tuning, Random Forest achieved an RMSE of \$1,903.31 and an R^2 of 0.5560, while Gradient Boosting achieved an RMSE of \$1,945.67 and an R^2 of 0.5361. These results showed that both ensemble methods improved upon the Linear Regression baseline, with Random Forest providing the strongest initial performance.

Hyperparameter tuning was then applied to the Gradient Boosting model to improve its predictive performance. After tuning, Gradient Boosting achieved an RMSE of \$1,884.82, an MAE of \$1,218.61, and an R^2 of 0.5646. Based on RMSE, the tuned Gradient Boosting model provided the best overall predictive performance among the models evaluated in this project. However, the analysis also showed that extreme high-cost claims remain more difficult to predict accurately because these claims represent only a very small portion of the dataset.

Based on these findings, the tuned Gradient Boosting model is most appropriate as a decision-support tool for ranking and prioritizing claims by predicted cost rather than for automatically setting claim reserves. Claims with higher predicted costs can be identified for earlier review, while experienced claims professionals remain responsible for evaluating individual cases and making final business decisions.

IV. METHODOLOGY / APPROACH

This project used 188,318 claims from the Allstate Claims Severity dataset.

Three models were tested: **Linear Regression**, **Random Forest**, and **Gradient Boosting**. After tuning, Gradient Boosting performed best with an RMSE of \$1,884.82.

The final model can help rank high-cost claims for early review.

V. DATA ANALYSIS

The analysis showed that 83.59% of claims are below \$5,000, while only 23 claims are above \$40,000. This shows that very expensive claims are rare.

Three regression models were compared: Linear Regression, Random Forest, and Gradient Boosting. After tuning, Gradient Boosting performed best, with an RMSE of \$1,884.82, an MAE of \$1,218.61, and an R^2 of 0.5646.

The final model can help rank potentially high-cost claims for early review.

A. *Key Dataset Attributes*

- 188,318 insurance claims
- 116 categorical features
- 14 numerical features
- Target variable: *loss* (claim cost)

B. *Analysis Focus Areas*

- Understand the distribution of claim costs
- Identify potentially high-cost claims
- Compare Linear Regression, Random Forest, and Gradient Boosting
- Select the best-performing model
- Use predictions to rank claims for early review

VI. EXPECTED OUTCOMES

The expected outcome of this project is a machine-learning model that can predict insurance claim costs and rank claims by their expected cost. The final tuned Gradient Boosting model can support claims managers by:

- Identifying potentially high-cost claims early.
- Ranking claims by predicted cost and flagging the top 5% for early review.
- Helping claims managers prioritize claims that may require additional attention.
- Supporting more informed reserve planning.
- Assisting claims professionals rather than making automatic reserve decisions.

In the held-out test set, the top 5% of claims ranked by the model captured all 6 claims with actual losses above \$40,000. This result provides encouraging evidence that the model can help identify rare, high-cost claims for early review. However, because the result is based on only 6 extreme claims, the 5% threshold should be validated on additional or future data before full operational deployment.

VII. TIMELINE

The project was completed over eight weeks. Table I summarizes the major activities completed during each stage.

TABLE I
PROJECT TIMELINE

Week	Activities
Weeks 1–2	Selected the project topic and dataset
Weeks 3–4	Cleaned the data and performed exploratory data analysis (EDA)
Week 5	Built and evaluated regression models
Week 6	Compared models and performed hyperparameter tuning
Week 7	Completed final model evaluation and dashboard development
Week 8	Completed the final report, presentation, and project submission

The major project tasks included data preparation, exploratory data analysis, model development, model comparison, hyperparameter tuning, final evaluation, dashboard development, and preparation of the final report and presentation.

VIII. CONCLUSION

This project demonstrated that machine learning can be used to predict insurance claim costs and rank claims by their expected cost. Among the models evaluated, the tuned Gradient Boosting model achieved the best overall performance, with an RMSE of \$1,884.82, an MAE of \$1,218.61, and an R^2 of 0.5646. Based on these results, Gradient Boosting was selected as the final model.

The final model is recommended as a decision-support tool to help claims managers identify potentially high-cost claims earlier and prioritize claims for review. As an initial pilot, the top 5% of model-ranked claims can be routed to experienced adjusters for early review before reserves are finalized. This approach may help claims managers prioritize resources and support more informed reserve planning.

In the held-out test set, the top 5% captured all 6 claims with actual losses above \$40,000. Although this result is encouraging, only 6 such extreme claims were present in the test set. Therefore, the 5% threshold should be validated on additional or future data before full operational deployment.

Overall, the model should be used to support claims professionals rather than replace their professional judgment. It is intended to improve claim prioritization and decision-making, not to automatically determine or set claim reserves.

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